Auto Scaling Groups.

1. What is ASG?

An **Auto Scaling Group (ASG)** in Amazon Web Services automatically manages a group of Amazon EC2 instances.

It ensures:

- The **right number of instances**
- At the **right time**
- Based on **demand**

2. Why ASG is Used

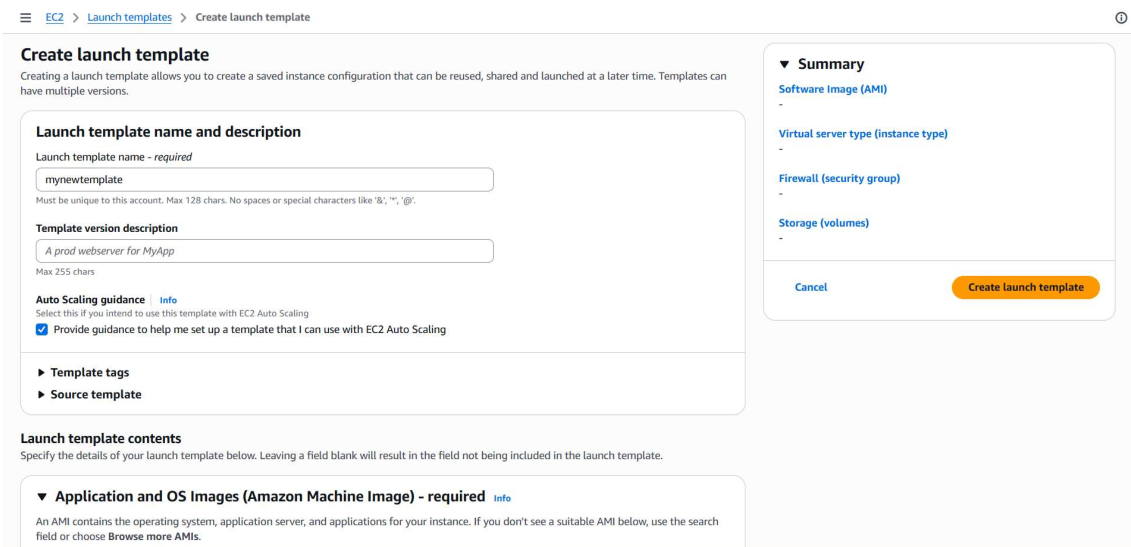
Without ASG:

- Manual scaling → slow + error-prone
- Over-provision → waste money
- Under-provision → system crash

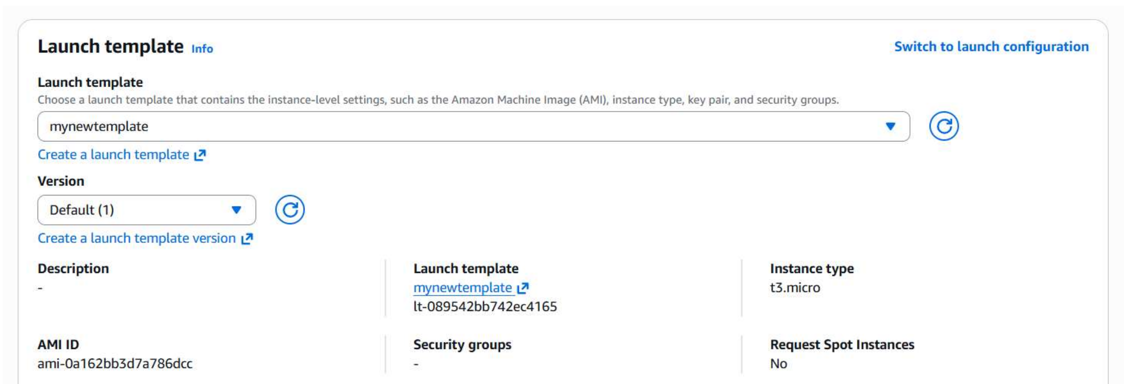
ASG solves all three:

- High availability
- Cost optimization
- Fault tolerance

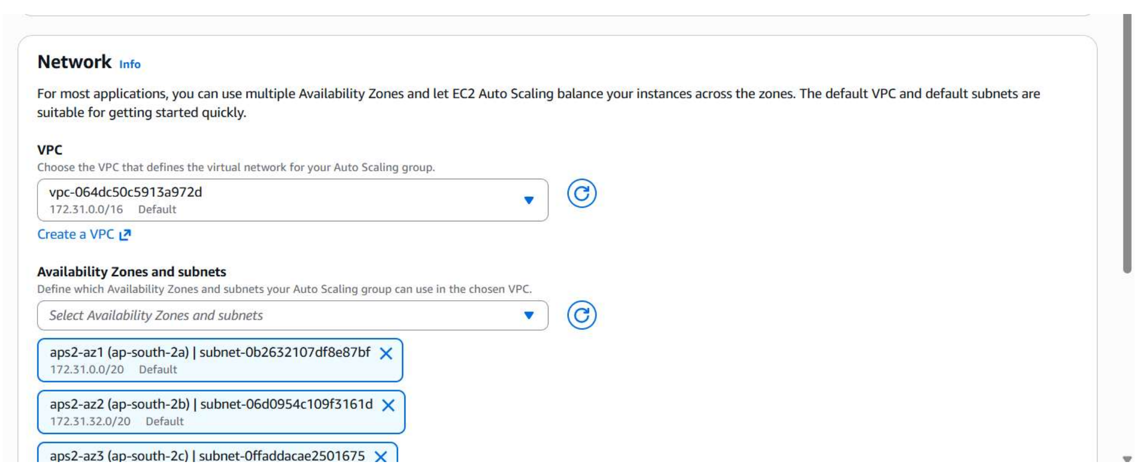
To create an ASG we have to create on Launch Template using the template the EC2 Creation will be easier. And ASG uses the Template to Maintain the Desired State.



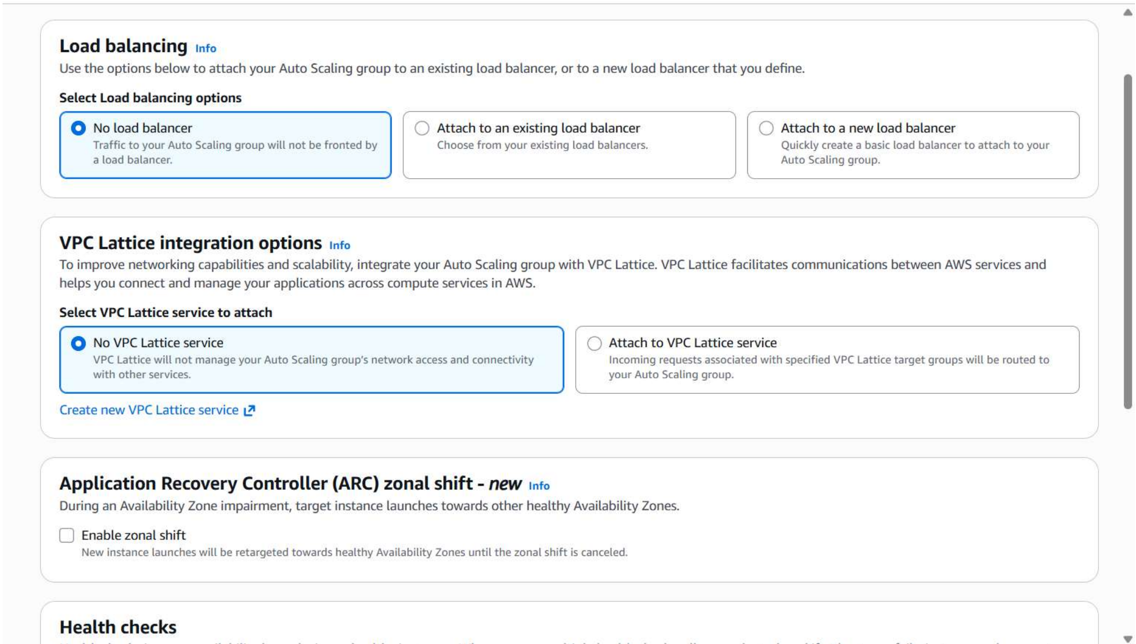
Then I want to create the ASG Using this Template.



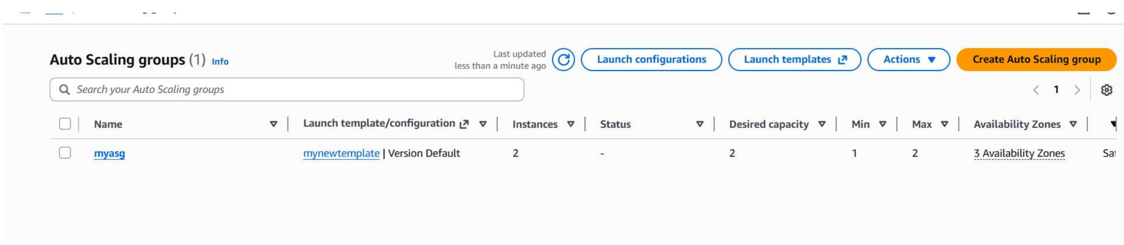
Then we have to configure network settings.



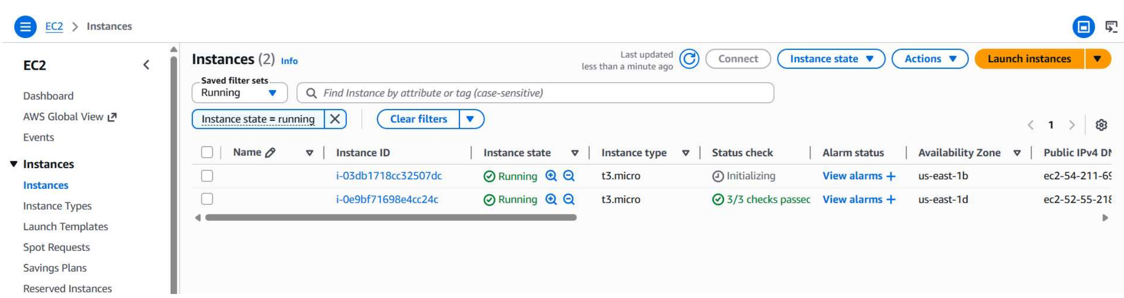
After that we have to select the with Load Balancer or without Load balancer.



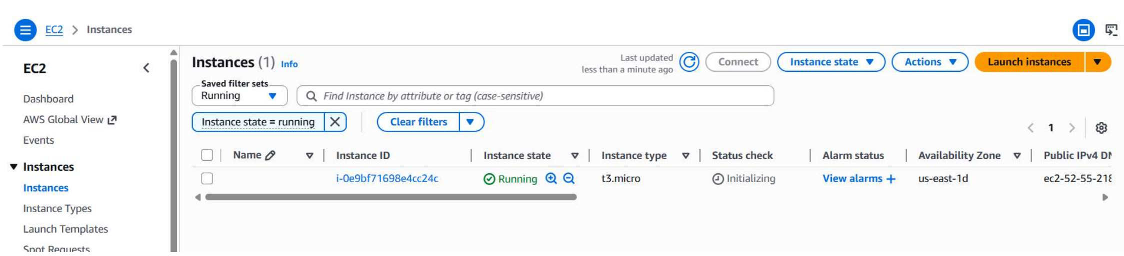
Created ASG.



After the ASG Creation the instances are automatically started creating using the ASG with the help of Launch Template.



It will recreate itself new Two Instances.

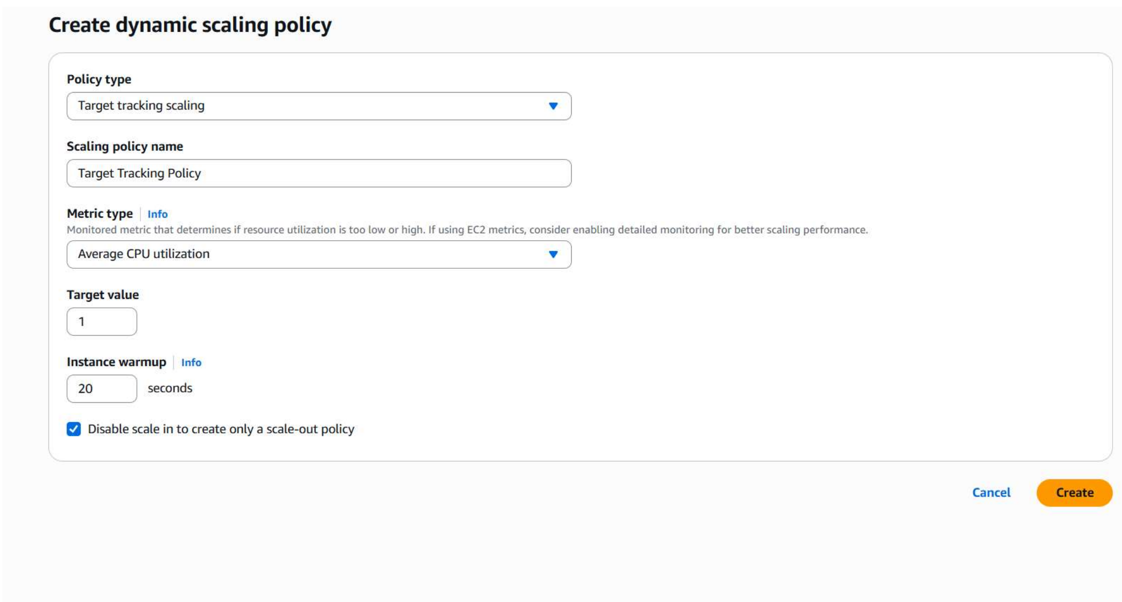


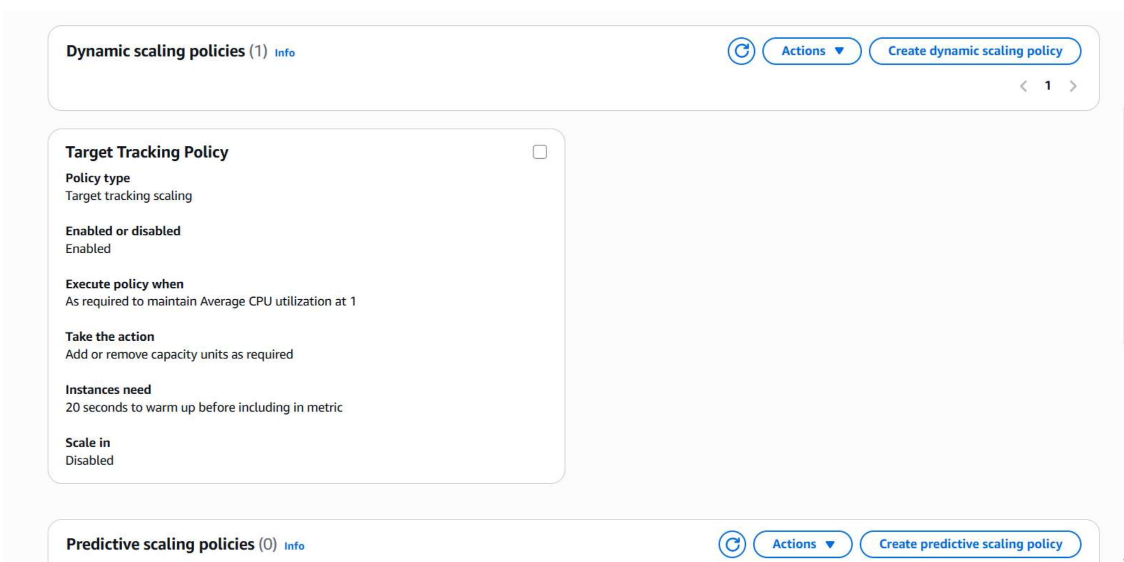
After we have to set some threshold to create an instance properly After utilizing the CPU of the instance.

Using the Dynamic Scaling Policies we can create the threshold for the ASG.

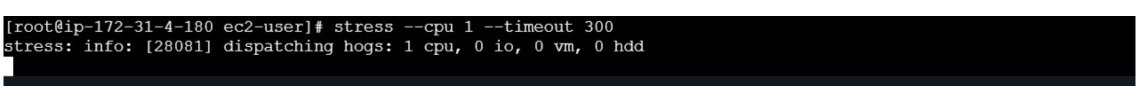
This is my Dynamic Policy with the 1% of CPU Utilization.

When the 1% Utilization is Hold Up For 5 mins It will Create One New Instance.





I have Give Fake Stress To Create The Instance.



This is the created Instance After the Creation Dynamic Scaling Policy.